
Elementary Geometry

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1 CLASSIC POINTS

Throughout this handout, we will reference the following points. It is important to be familiar with them.

- The incenter is the intersection of the angle bisectors. Usually, we denote this with I .
- The centroid is the intersection of the medians (lines bisecting the opposing side). Usually, we denote this with G .
- The orthocenter is the intersection of the altitudes. Usually, we denote this with H .
- The circumcenter is the intersection of the perpendicular bisectors. Usually, we denote this with O .
- Sometimes, we use (ABC) to denote the circumcircle; that is, the circle passing through A , B , and C .

In the future, we will show these points exist. That is, the angle bisectors concur at a single point, etc. For now, trust me.

2 ANGLE CHASING

The idea here is quite literally find the measures of angles. When you don't know what to do, a very good idea is to angle chase. This will reveal possibly similar triangles, concyclic points, and others you didn't see before. It may also point you in the direction and help you identify the main claims in the problem. Usually, olympiad geometry requires you to go through several claims to get the final answer. Angle chasing can definitely help you identify what you need to prove. There are a few notable things to remember before we dive deeper.

Let $\triangle ABC$ be a triangle with orthocenter H . Then,

$$\begin{aligned}\angle BHC &= 180^\circ - \angle HBC - \angle HCB \\ &= 180^\circ - (90^\circ - \angle C) - (90^\circ - \angle B) \\ &= \angle C + \angle B \\ &= 180^\circ - \angle A.\end{aligned}$$

This is a fascinating property as then, the reflection of H over BC lies on (ABC) . This is due to the fact that the opposite angles sum to 180° , a property we will explore later.

Let $\triangle ABC$ be a triangle with incenter I . Then,

$$\angle BIC = 180 - \angle IBC - \angle ICB = 180 - \frac{\angle B}{2} - \frac{\angle C}{2} = 90 + \frac{\angle A}{2}.$$

Then, if $\angle A = 60^\circ$, then $\angle BOC = 120^\circ$ since $\angle BOC = 2\angle A$, then $\angle BHC = 180^\circ - \angle A = 120^\circ$, and $\angle BIC = 120^\circ$. Therefore,

$$\angle BOC = \angle BIC = \angle BHC = 120^\circ.$$

So, the reflections of O and I over BC lies on (ABC) .

3 POWER OF POINT

Let P be a point and ω be a circle with center O and radius r . We define the power of a point P with respect to a circle ω as $\text{Pow}(P, \omega) = OP^2 - r^2$. A corollary from this is the famous power of point theorem.

- Suppose P lies outside a circle ω . Let A, B, C, D , and E are points on ω such that P, A, B and P, C, D are triplets of collinear points and PE is tangent to ω . Then:

$$PA \cdot PB = PC \cdot PD = PE^2.$$

- Suppose P lies inside a circle ω . Let A, B, C , and D are points on ω such that P, A, B and P, C, D are triplets of collinear points. Then:

$$PA \cdot PB = PC \cdot PD.$$

The *radical axis* of ω_1 and ω_2 is the line ℓ such that if $P \in \ell$, then $\text{Pow}(P, \omega_1) = \text{Pow}(P, \omega_2)$. Notably, if ω_1 and ω_2 intersect, then the radical axis is the line passing through the intersections. Crucially, the *radical center* is the intersection of the three pairwise radical axes of ω_1 , ω_2 , and ω_3 . This is notable in olympiads. If you wish to show three lines concur, you may want to use the above radical center to prove this.

A brief point: cyclic quadrilaterals are those whose four points are concyclic. Concyclic simply means lying on a circle. It is important to be familiar with both terms.

Say you wish to show $ABCD$ is cyclic. Then, there are a few ways we can go about this. (These aren't the only ways, but are common and simple ways to go about such.)

- Prove opposite angles sum to 180° . That is, $\angle ABC + \angle CDA = 180^\circ$.
- Show two angles subtend the same angle. That is, $\angle ABD = \angle ACD$.
- Use the converse of power of point.

4 INCENTER-EXCENTER LEMMA

Let $\triangle ABC$ be a triangle with incenter I . Suppose M is the midpoint of $\widehat{BC}$, not containing A . Then,

$$MB = MI = MC.$$

That is, M is the center of (BIC) . This isn't too hard to prove. Here is a walkthrough:

- M lies on AI because $\widehat{BM} = \widehat{MC}$.
- Show $\angle MIB = \angle MBI$ and $\angle MIC = \angle MCI$.

5 EULER LINE

The Euler Line is the line passing through H , G , and O . This is shocking, but not very obvious.

We will use a concept called *homothety*. We won't go too in-depth about what this means here. Here is a walk-through of the proof.

- Consider the medial triangle; that is, the triangle formed by the midpoints of $\triangle ABC$.
- Say the midpoints of BC , CA , and AB are M , N , and P , respectively.
- Notice, O is the orthocenter of the medial triangle.
- Here, we use an important theorem in relation to homothety. If $\triangle ABC$ and $\triangle XYZ$ are non-congruent, but their respective sides are parallel, then AX , BY , and CZ concur at the center of homothety.

- There exists homothety sending the medial triangle to $\triangle ABC$. The center of the homothety is the intersection of all lines formed by a point and the transformed point. That is, the intersection of AM , BN , and CP . This is just the medians, so the center of homothety is the centroid!
- Then, the centroid transforms $O \mapsto H$ as they are the orthocenters of their respective triangles. Thus, G lies on OH yielding the desired result.

This yields some very nice corollaries as G splits the median into a ratio of 1 : 2. Those include,

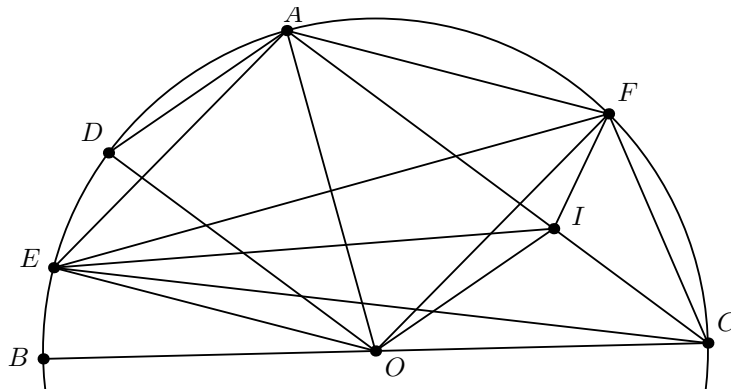
$$AH = 2 \cdot OM = 2R \cos(A).$$

In addition, $HG = 2 \cdot GO$.

6 EXAMPLE

2002 ISL G3. The circle S has centre O , and BC is a diameter of S . Let A be a point of S such that $\angle AOB < 120^\circ$. Let D be the midpoint of the arc AB which does not contain C . The line through O parallel to DA meets the line AC at I . The perpendicular bisector of OA meets S at E and at F . Prove that I is the incentre of the triangle CEF .

Proof.



Note that A is on the perpendicular bisector of $\angle FE$, so AC bisects $\angle ECF$.

In addition, as FE is the perpendicular bisector and it passes through O , then $OEAF$ is a rhombus. Suppose that $K = AD \cap BC$. Then, $AK \parallel OI$ implies $\triangle AKC \sim \triangle IOC$. So:

$$\begin{aligned} \angle AIO &= 180 - \angle OIC \\ &= 180 - \angle KAC \\ &= 180 - \angle DBK \\ &= \angle DBO \\ &= \angle DAO \\ &= \angle AOI. \end{aligned}$$

Thus, $AF = AO = AI = AE$.

By the Incenter-Excenter Lemma, then I is the incenter of $\triangle CEF$. ■

7 PROBLEMS

Problem 1. (*Classic*)

- (a) Prove AO and AH are isogonal lines; that is, the reflection of AO over the angle bisector of $\angle A$ is AH .
- (b) Prove that the reflection of H over the midpoint of BC is the antipode of A ; that is, diametrically opposite of A .
- (c) Quadrilateral $ABCD$ has midpoints W, X, Y , and Z of AB, BC, CD, DA . Characterize $WXYZ$.
- (d) (AoPS) Let $ABCDEF$ be a convex hexagon. Let A', B', C', D', E', F' be the centroids of triangles $FAB, ABC, BCD, CDE, DEF, EFA$, respectively. Show that every pair of opposite sides in hexagon $A'B'C'D'E'F'$ are parallel and equal in length.

Problem 2. (*Miguel Theorem*) Let ABC be a triangle, and D, E , and F be points on BC, CA , and AB , respectively. Show that the circumcircles of AEF, BFD , and CDE concur at a point, called the *Miguel Point*.

Problem 3. (*Nine-Point Circle*) Let ABC be a triangle. Show that the midpoints of AH, BH, CH, AB, BC , and CA lie on the circumference of a circle, along with the feet of the perpendiculars from A to BC, B to CA , and C to AB . In particular, the center is the midpoint of OH .

Problem 4. (*2012 British MO*) The point P lies inside triangle ABC so that $\angle ABP = \angle PCA$. The point Q is such that $PBQC$ is a parallelogram. Prove that $\angle QAB = \angle CAP$.

Problem 5. (*1995 IMO*) Points A, B, C, D lie on the line ℓ in this order. Circles ω_C, ω_B with diameters respectively AC, BD intersect at points X, Y . Point P lies on the line XY , but not on ℓ . Denote by M, N the intersection of respectively CP with ω_C and BP with ω_B . Prove that the lines AM, DN, XY are concurrent.

Problem 6. (*2025 HMMT*) Let ω_1 and ω_2 be two circles intersecting at distinct points A and B . Point X varies along ω_1 , and point Y is chosen on ω_2 such that AB bisects angle $\angle XAY$. Prove that as X varies along ω_1 , the circumcenter of $\triangle XAY$ (if it exists) varies along a fixed line.

Problem 7. (*2023 USA EGMO TST*) Let ABC be a triangle with $AB + AC = 3BC$. The B -excircle touches side AC and line BC at E and D , respectively. The C -excircle touches side AB at F . Let lines CF and DE meet at P . Prove that $\angle PBC = 90^\circ$.

Problem 8. (*2024 APMO*) Let ABC be an acute triangle. Let D be a point on side AB and E be a point on side AC such that lines BC and DE are parallel. Let X be an interior point of $BCED$. Suppose rays DX and EX meet side BC at points P and Q , respectively, such that both P and Q lie between B and C . Suppose that the circumcircles of triangles BQX and CPX intersect at a point $Y \neq X$. Prove that the points A, X , and Y are collinear.

Problem 9. (*2019 USEMO*) Let $ABCD$ be a cyclic quadrilateral. A circle centered at O passes through B and D and meets lines BA and BC again at points E and F (distinct from A, B, C). Let H denote the orthocenter of triangle DEF . Prove that if lines AC, DO, EF are concurrent, then triangle ABC and EHF are similar.

Problem 10. (*2024 ISL*) Let $ABCD$ be a cyclic quadrilateral such that $AC < BD < AD$ and $\angle DBA < 90^\circ$. Point E lies on the line through D parallel to AB such that E and C lie on opposite sides of line AD , and $AC = DE$. Point F lies on the line through A parallel to CD such that F and C lie on opposite sides of line AD , and $BD = AF$. Prove that the perpendicular bisectors of segments BC and EF intersect on the circumcircle of $ABCD$.

8 SOLUTIONS

8.1 CLASSIC

Prove all parts:

- (a) Prove AO and AH are isogonal lines; that is, the reflection of AO over the angle bisector of $\angle A$ is AH .
- (b) Prove that the reflection of H over the midpoint of BC is the antipode of A ; that is, diametrically opposite of A .
- (c) Quadrilateral $ABCD$ has midpoints W, X, Y , and Z of AB, BC, CD, DA . Characterize $WXYZ$.
- (d) Let $ABCDEF$ be a convex hexagon. Let A', B', C', D', E', F' be the centroids of triangles $FAB, ABC, BCD, CDE, DEF, EFA$, respectively. Show that every pair of opposite sides in hexagon $A'B'C'D'E'F'$ are parallel and equal in length.

Solution.

- (a) We angle chase. Clearly, $\angle BAH = 90^\circ - \angle B$. Now,

$$\angle CAO = \frac{180^\circ - \angle AOC}{2} = \frac{180^\circ - 2\angle B}{2} = 90^\circ - \angle B.$$

Thus, $\angle BAH = \angle CAO$. Hence, AO and AH are reflections, as desired.

- (b) Let H' and H'' be the reflection of H over BC and the midpoint of BC , respectively. Recall, $\angle BHC = 180^\circ - \angle A$. Then, $BHH'C$ is a parallelogram, so $\angle BH'C = 180^\circ - \angle A$. So, $BH'CA$ is cyclic. Notice, $H'H'' \perp AH$, so $\angle AH'H'' = 90^\circ$. Thus, AH'' is the diameter, as desired.

- (c) The mid-line of a triangle is parallel to the corresponding side. Notice, WX is the midline of $\triangle BAC$, so $WX \parallel AC$. Similarly, $YZ \parallel AC$, so $WX \parallel YZ$. Hence, opposite sides are parallel, and thus $WXYZ$ is a parallelogram.

- (d) Let M be the midpoint of AB . Then,

$$\frac{MA'}{MF} = \frac{MB'}{MC} = \frac{1}{3}.$$

Hence, $\triangle MA'B' \sim \triangle MFC$ and $A'B' \parallel FC$, and $A'B' = \frac{1}{3}FC$. Similarly, $D'E' \parallel FC$ and $D'E' = \frac{1}{3}FC$.

Thus, $A'B' \parallel D'E'$ and $A'B' = D'E'$. Thus, opposite sides are parallel and equal in length, as desired.

8.2 MIGUEL POINT

Let ABC be a triangle, and D , E , and F be points on BC , CA , and AB , respectively. Show that the circumcircles of AEF , BFD , and CDE concur at a point, called the *Miguel Point*.

Solution. Let $P \neq D$ be the second intersection of (AEF) and (BFD) . We will show that P lies on (CDE) .

Notice,

$$\begin{aligned}\angle CDP + \angle CEP &= (180 - \angle BDP) + (180 - \angle AEP) \\ &= \angle BFP + \angle AFP \\ &= 180^\circ.\end{aligned}$$

Hence, $CDPE$ is cyclic, as desired.

8.3 NINE-POINT CIRCLE

Let ABC be a triangle. Show that the midpoints of AH , BH , CH , AB , BC , and CA lie on the circumference of a circle, along with the feet of the perpendiculars from A to BC , B to CA , and C to AB . In particular, the center is the midpoint of OH .

Solution. Let M , N , and P be the midpoints of AB , BC , and CA . Let X , Y , and Z be the midpoints of AH , BH , and CH . Let D , E , and F be the feet of the perpendiculars from A to BC , B to CA , and C to AB .

We will show X , Y , and Z lie on (MNP) . Notice, there exists homothety from $\triangle APM \mapsto \triangle ABC$ which maps $X \mapsto H$, with scale factor 2. Thus,

$$\angle PXN = \angle BHC = 180^\circ - \angle A.$$

Hence, $\angle PXM + \angle PMN = (180^\circ - \angle A) + \angle A = 180^\circ$. Hence, X lies on (MNP) . Likewise, Y and Z both lie on (MNP) .

Now, we will show D , E , and F lie on (XYZ) . Then, $\triangle XYZ \sim \triangle ABC$ due to homothety with scale factor 2. Notice, $HX : XA = 1$, so E and H are reflections over XZ . Thus,

$$\begin{aligned} \angle XEZ + \angle XYZ &= \angle AHC + \angle ABC \\ &= (180^\circ - \angle B) + \angle B \\ &= 180^\circ. \end{aligned}$$

Hence, D , E , and F lie on (XYZ) . Thus, $\{M, N, P, X, Y, Z, D, E, F\}$ all lie on a circle.

Recall, $AH = 2 \cdot OM$. Since X is the midpoint of AH , then $AX = XH = OM$. As $AH \parallel OM$, then $XHMO$ is a parallelogram. Since $\angle XDM = 90^\circ$, then the midpoint of XM is the center of the nine-point circle. As $XHMO$ is a parallelogram, then the midpoint of XM is the midpoint of OH , as desired.

This is an amazing conclusion. Recall, H , G , O are collinear with ratio $HG : GO = 2 : 1$!

8.4 2012 BRMO

The point P lies inside triangle ABC so that $\angle ABP = \angle PCA$. The point Q is such that $PBQC$ is a parallelogram. Prove that $\angle QAB = \angle CAP$.

Solution. Let AB intersect PC at X and AC intersect BP at Y . Then, $BXYC$ is cyclic. Also, $PBQC$ is a parallelogram. So:

$$\begin{aligned}\angle AXP &= \angle AXY + \angle YXP \\ &= \angle ACB + \angle YBC \\ &= \angle ACB + \angle BCQ \\ &= \angle ACQ.\end{aligned}$$

Similarly, $\angle AYP = \angle ABQ$. Thus, $AXPY \sim ACQB$. Hence, $\angle CAP = \angle BAQ$, as desired.

8.5 1995 IMO

Points A, B, C, D lie on the line ℓ in this order. Circles ω_C, ω_B with diameters respectively AC, BD intersect at points X, Y . Point P lies on the line XY , but not on ℓ . Denote by M, N the intersection of respectively CP with ω_C and BP with ω_B . Prove that the lines AM, DN, XY are concurrent.

Solution. First, due to power of point at P ,

$$CP \cdot PM = XP \cdot PY = BP \cdot PN.$$

Hence, $BCNM$ is cyclic.

Then,

$$\begin{aligned} \angle MAD + \angle MND &= \angle MAD + (\angle MNB + \angle BND) \\ &= (\angle MAD + \angle MCB) + \angle BND \\ &= 180^\circ. \end{aligned}$$

Hence, $AMND$ is cyclic.

The pairwise radical axes of (AMC) , (BND) , and $(AMND)$ is AM, DN , and XY . Thus, these lines concur at the radical center, as desired.

8.6 2025 HMMT

Let ω_1 and ω_2 be two circles intersecting at distinct points A and B . Point X varies along ω_1 , and point Y is chosen on ω_2 such that AB bisects angle $\angle XAY$. Prove that as X varies along ω_1 , the circumcenter of $\triangle AXY$ (if it exists) varies along a fixed line.

Solution. Let M and N be midpoints of AX and AY , and let P be the midpoint of AB . Note that AMO_1P and ANO_2P are cyclic. Then:

$$\angle OO_1O_2 = 180 - \angle MO_1O_2 = 180 - \angle MAP = 180 - \angle NAP = \angle NO_2O_1 = \angle O_1O_2O.$$

Hence, O lies on the perpendicular bisector of O_1O_2 .

8.7 2023 USA EGMO TST

Let ABC be a triangle with $AB + AC = 3BC$. The B -excircle touches side AC and line BC at E and D , respectively. The C -excircle touches side AB at F . Let lines CF and DE meet at P . Prove that $\angle PBC = 90^\circ$.

Solution. Let P and Q be points of tangency of the incircle and AB and BC , respectively. Let G be the point of tangency with the C -excircle and BC . Recall, the reflection of the point of tangency between the incircle and the side across the midpoint is the point of tangency between the excircle and the side. Hence:

$$GB = FB = AP = BC = AQ = EC = CD.$$

Denote I as the incenter, then:

$$\angle ICB = \frac{1}{2} \cdot \angle ECB = \frac{1}{2} \cdot (180^\circ - \angle ECD) = \angle EDC.$$

Hence, $PD \parallel IC$.

Denote I_C as the excenter of the C -excircle. Then, as $FB = FC$:

$$\angle FCB = \frac{1}{2} \cdot (180^\circ - \angle FBC) = \frac{1}{2} \cdot \angle FBG = \angle I_C B G.$$

Thus, $I_C B \parallel PC$.

Notice, $GB = BC = CD$. Upon shifting $PD \mapsto I_C C$ and $PC \mapsto I_C B$, then $\triangle PBD \cong \triangle I_C GC$. Therefore, $\angle PBC = \angle I_C GC = 90^\circ$, as desired.

8.8 2024 APMO

Let ABC be an acute triangle. Let D be a point on side AB and E be a point on side AC such that lines BC and DE are parallel. Let X be an interior point of $BCED$. Suppose rays DX and EX meet side BC at points P and Q , respectively, such that both P and Q lie between B and C . Suppose that the circumcircles of triangles BQX and CPX intersect at a point $Y \neq X$. Prove that the points A, X , and Y are collinear.

Solution. Let $\triangle DXE$ intersect AB and AC at Y and Z , respectively. Then,

$$\begin{aligned}\angle XZC &= \angle XPC = \angle XPQ = \angle XDE, \\ \angle XYD &= \angle XQB = \angle XQP = \angle XED.\end{aligned}$$

Therefore, Y and Z lie on (DEX) . Thus, by power of point at A , we find:

$$AD \cdot AY = AZ \cdot AE \implies \frac{AY}{AZ} = \frac{AE}{AD} = \frac{AC}{AB} \implies AY \cdot AB = AZ \cdot AC.$$

Thus, A lies on the radical axis of XY , as desired.

8.9 2019 USEMO

Let $ABCD$ be a cyclic quadrilateral. A circle centered at O passes through B and D and meets lines BA and BC again at points E and F (distinct from A, B, C). Let H denote the orthocenter of triangle DEF . Prove that if lines AC, DO, EF are concurrent, then triangle ABC and EHF are similar.

Solution. Let $P = AC \cap EF$. The condition implies $\angle BCD = \angle EAD$ and $\angle CFD = \angle AED$. Hence, $\triangle FCD \sim \triangle EAD$. Therefore, $\angle FDC = \angle EDA$. So,

$$\angle ABC = 180 - \angle ADC = 180 - \angle FDE = \angle EHF.$$

Angle chasing,

$$\angle DEF = \angle DBF = \angle DAC.$$

Hence, $EAPD$ is cyclic. Now, consider:

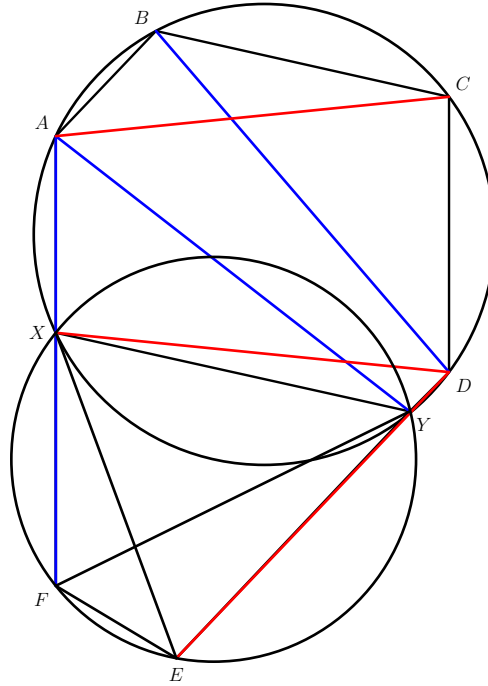
$$\angle BAC = 180 - \angle CAE = \angle EDP = \angle EDO = \angle HDF = \angle HEF.$$

The fourth equality comes from O and H isogonality in $\triangle DEF$. Hence, $\triangle ABC \sim \triangle EHF$, as desired.

Let $ABCD$ be a cyclic quadrilateral such that $AC < BD < AD$ and $\angle DBA < 90^\circ$. Point E lies on the line through D parallel to AB such that E and C lie on opposite sides of line AD , and $AC = DE$. Point F lies on the line through A parallel to CD such that F and C lie on opposite sides of line AD , and $BD = AF$.

Prove that the perpendicular bisectors of segments BC and EF intersect on the circumcircle of $ABCD$.

Solution.



Let the line passing through A parallel to CD and the line through D parallel to AB intersect $(ABCD)$ at X and Y , respectively.

Then, $CDXA$ and $ABDY$ are isosceles trapezoids, so $DX = AC = DE$ and $AY = BD = AF$. Therefore, $\triangle DEX$ and $\triangle AFY$ are isosceles triangles. But,

$$\angle XDE = \angle XDY = \angle XAY = \angle FAY.$$

So, we may conclude

$$\angle XFY = \angle AFY = \frac{180 - \angle FAY}{2} = \frac{180 - \angle XDE}{2} = \angle XED = \angle XEY.$$

Therefore, we may conclude XFY is cyclic.

Now, $AB \parallel DY$ and $AX \parallel CD$, so $\angle BAX = \angle CDY$. Hence:

$$\angle XYB = 180 - \angle XAB = 180 - \angle YDC = \angle YBC.$$

Hence, $BC \parallel XY$. As both BC and XY are chords of $(ABCD)$, then the perpendicular bisectors of BC and XY are equivalent. It suffices to show the perpendicular bisectors of XY and FE intersect on $(ABCD)$. But,

$XYEF$ is a cyclic quadrilateral. thus, it suffices to show the circumcenter of $(XYEF)$ lies on $(ABCD)$.

Let O be the circumcenter of $(XYEF)$. Then, as $\triangle FAY$ is isosceles.

$$\angle XOY + \angle XAY = 2\angle XFY + \angle XAY = \angle AFY + \angle AYF + \angle FAY = 180^\circ.$$

Hence, O lies on $(ABCD)$ and the perpendicular bisectors of BC and EF intersect on $(ABCD)$, as desired.